CSCA-MVM 22 October 1991

FOR: Record

SUBJECT: Description of the Map Grid Used in the Ardennes Data Base

1. **Background.** This memo gives a rather complete description of the map grid system used in the Ardennes Combat Simulation Data Base (ACSDB). It is intended to be the last of this series, in the sense that it replaces, expands upon, and/or supersedes all my previous memos on this subject. This series of memos was made necessary by the following considerations.

a. The ACSDB reports received from Data Management Systems Incorporated (DMSI) describe the map grid system used as follows (see the part of their definition of the HQ Starting Location Field, which is Field 5 of the Unit Location part of the ACSDB, as presented, for example, in Section IV of Volume 2 of DMSI's ACSDB report):

"HQ Starting Location; Field 5. Recorded under 'HQ Starting Location is the location of the unit's headquarters at the beginning of the day. Its location is designated in all cases by the Universal Transverse Mercator (UTM) four- or six-digit grid coordinates used on the 1943 Geographical Section, General Staff maps produced by the Army Map Service. These maps are:

Sheet#	Name
8	Brussels
9	Liege
12	Namur
13	Marche
16	Mezieres
14	Arlon
R-1	Koeln
S-1	Bonn
T-1	Trier
V-1	Neunkirchen

If information is available, an actual place name is included, most commonly a town, village, or city name."

b. Unfortunately, this characterization of the maps is inaccurate. The description on pg xiii of Hugh M. Cole's The Ardennes: Battle of the Bulge, a volume of the "Green Book" series on the United States Army in World War II and first printed in 1965, is closer to the truth: "The maps consulted by the author were those in use at the end of 1944. They include the U.S. Army reproductions of the maps

prepared by the British Geographical Section, General Staff, in the 1:25,000 series (G.S., G.S. 4041), the 1:50,000 series (G.S., G.S. 4040), and the 1:100,000 series (G.S., G.S. 4336 and 4416)." That is, the maps, whether technically supplied by the British or by the US, were either originals or direct copies of the ***British*** maps of the Ardennes area. Now, those British maps ***do not*** use the UTM system. Instead, their grid references actually are the four- or six-digit grid references used in the British "Nord de Guerre Zone," and based on the Lambert Conformal Conical Projection (or, as the British call it, the Lambert Conical Orthomorphic Projection), as shown at ENCL 1, which was extracted from AMS Technical Manual No. 36, Grids and Grid References, Army Map Service, Corps of Engineers, Dept. of the Army, January 1950.

c. DMSI's ACSDB report, Volume I, Section XII, states that "One of the deliverables for the Ardennes Campaign Simulation Data Base (ACSDB) is a clear acetate overlay showing positions of Allied and German units at the beginning of the Ardennes Campaign on 16 December 1944. ... The maps used with the overlays are the same maps used to record Unit Location and Air Data in the ACSDB. They are ten 1943 Geographical Section, General Staff, maps produced by the Army Map Service. Full-scale color photographs of these maps were provided to the US Army Concepts Analysis Agency." Unfortunately, a search of CAA's holdings has not located either the full-scale color photographs of these maps or the acetate overlays. Accordingly, the conversion of ACSDB grid references to latitude and longitude has to be recreated.

d. Since the ACSDB grid references are not related to the US Army's current UTM-style grid references, CAA needs some way to convert them to latitudes and longitudes, or to the current UTM grid references. As the Concept Evaluation Model's (CEM's) FEBA trace outputs are commonly expressed in latitudes and longitudes, it is perhaps most desirable to have a system for converting the ACSDB grid references to latitudes and longitudes. Because we know the details of the British map projection system (see ENCL 1), we could (at least in principle) use its mathematical properties to perform the conversion. So doing would, no doubt, provide the greatest accuracy (in fact, far greater accuracy than could possibly be justified by the approximate nature of the location data in the ACSDB, which typically is expressed to the nearest km and never to better than the nearest 0.1km). But such an approach is not very practical, for it would involve extensive research into esoteric cartographic properties of the British Lambert Conical Orthomorphic (modified) system, as well as the characteristics of the Du Plessis (reconstituted) spheroid. This would be a lot of trouble, likely more than its accuracy is worth in practice, and wouldn't lead to simple and easily-used conversion algorithms. Accordingly, we adopted a much easier approach that uses only information and methods readily available to CAA and which is easily implemented. It is described in detail in para 6, below.

e. We also have the REFLEX® files provided by DMSI, which purport to represent the unit locations. These files can be used to plot the locations of units on a Cartesian coordinate system. However, the relation of this Cartesian coordinate system to latitude and longitude is not known. We would like to know where the origin of this REFLEX® system is located, what unit of measurement is used for its scales, and how its axes are oriented relative to latitude and longitude.

f. Accordingly, CAA is faced with the following questions. Each of them is answered in the following paragraphs of this memo.

(1) What are all of the map sheets and grid references actually used in the ACSDB data files?

(2) What scale is used on these map sheets? What do the ACSDB grid references signify?

(3) How do the map sheets designated in the ACSDB entries relate to one another? That is, does map sheet U adjoin map sheet L? And, if so, is map sheet U north, south, east, or west of map sheet L?

(4) How does one convert these map sheet coordinates to and from those used in the REFLEX® plot files?

(5) How does one convert from these ACSDB map sheet coordinates to and from latitudes and longitudes, or to and from the current UTM map system coordinates?

(6) Are there any known errors in the locations reported in the ACSDB? If so, what are they?

2. What Grid References Are Used.

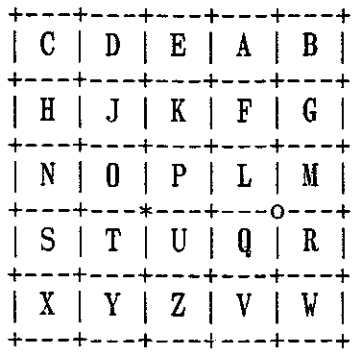
a. The HQ Start Location Records. So far, we have looked mainly at the locations recorded in the ACSDB's HQ Starting Location records, that is, in Field 5 of the Unit Location file. This was done because they provide a manageable sample of ACSDB's locations. The following is thought to be an exhaustive tabulation of all of the forms of entry used for that field. In the tabulation below, a symbol of the form Z---- stands for map sheet alphabetical designation Z, followed by either a four- or six-digit grid coordinate, followed in turn by either (i) blank spaces, (ii) a blank space and a place name or comment (usually in parentheses), (iii) a place name or comment (usually in parentheses) with no blanks between it and the grid reference, or (iv) a dash (-) and a second alphanumeric grid reference.

Entry Form	Remarks
	This is an entry consisting entirely of blanks.
(BOCKANGE)	The name of a location, in parentheses.
(EISENSCHMIDT)	ditto
(ENNERY)	ditto
APACH, FRANCE	The name of a location, not in parentheses.
NANCY, FRANCE	ditto
Rodemack, France	ditto
(MOVING)	Written out, in parentheses.
In transit	Written out, not in parentheses.
Vic L----	Location on map sheet L, preceded by "Vic."
Vic P----	ditto
F----	
J----	
K----	
L----	
0----	Map sheet alfa "Oh".
P----	
Q----	
T----	There are only two entries for this map sheet.
U----	
k----	There is only one such entry, and it is obviously a typo for K----.

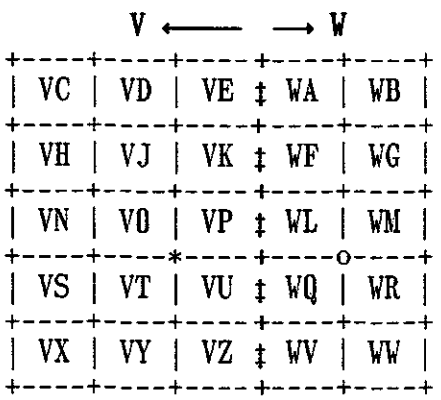
b. Other Data Fields. Location information is also provided in other ACSDB fields. These either mark the left and right flank of units deployed on line or else give various "reference points" describing such things as the twists and turns of the front line, or the outline of the "goose egg" unit location for units not on line, or the route taken by units in motion, and so forth. We did not attempt to make an exhaustive tabulation of all such entries. Some fields are blank and some are all question marks (i.e., ?????) or dashes (----), indicating that data for that field is not available. A hasty scan of a few sample cases indicates that the grid references in these other fields are well-represented by that in the HQ Start Location field—that is, the HQ Location Field provides an adequate sample of the kinds of grid references recorded in the other fields.

3. Map Scale. We make the following comments regarding the map scale used. In the first place, if only relative locations were needed, then the actual map scale would be largely irrelevant. However, for some purposes, it is necessary to know absolute locations. The fact is that each map sheet covers an area 100km by 100km. Hence, for example, the entry P1245 designates a point 12km east and 45km north of the 0000-point on map sheet P, where east and north are the grid east and grid north and so may not coincide with the geographical lines of latitude and longitude. Similarly, P123456 designates a point 12.3km (grid) east and 45.6km (grid) north of the origin point of map sheet P.

4. **How the Map Sheets Fit Together.** We make the following comments regarding how the map sheets fit together. Originally, we were able to determine this by comparing the ACSDB coordinates to locations shown on Plate 61 (Belgium:Luxembourg) and Plate 64 (Germany, South) of the Times Atlas of the World, using the place names cited in the ACSDB to provide the linkage between the two. Subsequently, we confirmed this against the Nord de Guerre map sheet labeling system. In this way we determined that the map sheets fit together as shown below:



where o marks the "False Origin" of the Nord du Guerre system (at 40°30′N and 7°44′13.95″E), and * marks the origin of the so-called UniGrid Cartesian coordinate system which I introduced to assist in various grid considerations (in particular, all grid system conversions will use the UniGrid system as the standard, that is, all conversions will be to and from, or via, the UniGrid system). The UniGrid coordinate system has its x-axis directed due (grid) east, and its y-axis directed due (grid) north. Each of these lettered map sheets measures 100km on a side. The part of the Nord de Guerre Zone used in the ACSDB includes parts of both 500,000 meter squares V and W. Thus, a more precise designation of the map grids used in the ACSDB is as follows, which can be related directly to the British Nord de Guerre Zone chart at ENCL 1:



We remark that all of the ACSDB grid references we have seen so far fall into the nine central map squares J, K, F, O, P, L, T, U, and Q. The bordering map squares were introduced above merely to illustrate how they would fit together with the nine central ACSDB map squares, and to make some provision for the possibility that ACSDB grid references in the bordering map squares might be encountered in the future.

5. Conversion To and From REFLEX® Coordinates. The REFLEX® coordinate system is identical to the UniGrid system defined in the preceding paragraph. In fact, the UniGrid coordinate system was chosen to agree with the REFLEX® coordinate system. Accordingly, conversions between the ACSDB and REFLEX® system are the same as for UniGrid system. We present the algorithm for conversions between the ACSDB and UniGrid systems.

a. From ACSDB to UniGrid. Let Ωuv be the ACSDB grid reference, where Ω is an alphabetical letter and u and v are grid coordinates. For example, for the ACSDB grid reference Q789123, $\Omega = Q$, $u = 78.9$km, and $v = 12.3$km. With this notation, the conversion from ACSDB grid references to the UniGrid Cartesian coordinate system is

$$\Omega uv \rightarrow \text{UniGrid}(u',v'), \text{ where}$$
$$u' = \text{EOffset}(\Omega) + u, \text{ and}$$
$$v' = \text{NOffset}(\Omega) + v.$$

The E(asting) and N(orthing) offsets, in km, are as given by the following table:

Ω	EOffset(Ω)	NOffset(Ω)
C	−200	+200
D	−100	+200
E	0	+200
A	+100	+200
B	+200	+200
H	−200	+100
J	−100	+100
K	0	+100
F	+100	+100
G	+200	+100
N	−200	0
O	−100	0
P	0	0
L	+100	0
M	+200	0

S	−200	−100
T	−100	−100
U	0	−100
Q	+100	−100
R	+200	−100
X	−200	−200
Y	−100	−200
Z	0	−200
V	+100	−200
W	+200	−200

For example, the conversion of ACSDB grid reference Q789123 proceeds as follows. Since $\Omega = Q$, $EOffset(\Omega) = +100$ and $NOffset(\Omega) = -100$. So the corresponding UniGrid coordinates are $u' = EOffset(\Omega) + u = +100 + 78.9 = 178.9km$, and $v' = NOffset(\Omega) + v = -100 + 12.3 = -87.7km$. A PowerBASIC version of this algorithm is given at ENCL 2.

b. From UniGrid to ACSDB. This is just the reverse of the above procedure. A PowerBASIC version of this algorithm is given at ENCL 3.

6. Conversion to Geographical Coordinates. We make the following comments regarding conversion of ACSDB map coordinates to geographical or current UTM coordinates. As mentioned in para 1d, above, an "exact" conversion system is probably not worth the trouble. Hence, a simpler method is used, as explained below.

a. First, to relate grid references and latitudes and longitudes, we rely upon the place names occasionally given along with the ACSDB grid references. The presumptions are that (i) neither their grid references nor their latitudes and longitudes have changed since WW II, that (ii) the ACSDB grid references are those of the place, rather than (or as well as) of the unit, and that (iii) their latitudes and longitudes are as recorded in the gazetteer section of the Times Atlas of the World. While it is clear that these assumptions may not hold universally[1], we believe that in general they are valid. Also, it is to some extent possible to check the validity of these assumptions, as will be explained later.

b. Second, we used the UniGrid system introduced in para 4 as the standard, so that all the *direct* conversions are to and from it. Thus, to convert ACSDB grid references to latitudes and longitudes, we first convert the ACSDB grid reference to its corresponding UniGrid coordinate, and finally convert that to latitude and longitude. And similarly for the reverse conversion.

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1 For instance, the unit's actual location may well have been near, but not actually *in*, the locality designated in the ACSDB. Also, there may have been some slight errors in recording the ACSDB coordinates. The upshot is that the ACSDB coordinates, even when a place name is cited, do not necessarily mean that the unit's location coincided exactly with that place's location.

c. The theory behind the proposed method is as follows. We assume that the geographical longitude and latitude coordinates are related to the UniGrid coordinates by a linear or quadratic expression, with some more-or-less random residual errors. Specifically, we assume that

$$y_1 = a_{10} + a_{11}x_1 + a_{12}x_2 + b_{111}{x_1}^2 + b_{112}x_1x_2 + b_{122}{x_2}^2 + \epsilon_1$$

$$y_2 = a_{20} + a_{21}x_1 + a_{22}x_2 + b_{211}{x_1}^2 + b_{212}x_1x_2 + b_{222}{x_2}^2 + \epsilon_2,$$

where y_1 is the longitude of the place (in degrees and decimal degrees east of the Greenwich Prime Meridian) and y_2 its latitude (in degrees and decimal degrees north of the Equator); x_1 and x_2 are the UniGrid easting and northing grid coordinates, respectively; and ϵ_1 and ϵ_2 are error terms. We make the following comments.

(1) In the quadratic equations for the conversion of coordinates there are a total of 12 unknown parameters whose values are to be determined. Each given location provides a pair of values. Hence, a minimum of 6 given locations is needed to completely solve for all 12 unknown parameters. If only the linear terms are retained, then there are a total of 6 unknown constants, and three given locations is the minimum required. In each case, additional given locations are desirable, and are needed if estimates of the error terms, ϵ, are to be made.

(2) To reduce the problems introduced by failure of the unit location to coincide with the place named in the ACSDB, we used only place names for which a single ACSDB grid reference was given, presuming it to be the location of the place named.

(3) The proposed functional form for relating longitude and latitude to UniGrid coordinates is linear in the parameters a_{ij} and b_{ijk}. Hence, conventional least-squares procedures can be used to fit these parameter values to the given data, which provides matches of the form: "the longitude and latitude pair (y_1,y_2) corresponds to the UniGrid point (x_1,x_2)." The values ϵ_1 and ϵ_2 provide an estimate of the error involved in making this correspondence. These ϵ error values *also* provide a check on gross errors in matching longitude and latitude values to UniGrid coordinates. (Many statistical packages provide for such least-squares fits through their regression programs. Regression programs providing robust regression calculations also provide for minimizing other functions of the errors, such as the sum of their absolute values, etc.)

(4) For portions of the earth's surface that are not too large, the use of a quadratic (or, indeed, even a linear) fit is probably satisfactory for most of CAA's purposes—that is, higher order terms need not be included. The principal error in any such fit is likely to be due to the (nonlinear)

"convergence of the meridians," that is, the meridians become closer together as one progresses further north. The following quick computation shows that, at least for the case in hand, even this error is not particularly large. The radius r of the locus of equal-latitude points at latitude λ is equal to

$$r = \rho \cos\lambda,$$

where ρ is the radius of the earth. Hence,

$$dr/r = - \tan\lambda \, d\lambda.$$

Evaluating this for $\lambda = 50^\circ$ latitude (a typical latitude for the ACSDB data) and $d\lambda = \pm\frac{1}{2}^\circ$ (since the ACSDB data span about a degree of latitude, or $\pm\frac{1}{2}$ a degree) yields

$$dr/r = \pm 0.0104,$$

so that in our case the error involved in ignoring the convergence of the meridians amounts to about ± 1 percent. Using a quadratic fit should pretty much completely eliminate this amount of error for all practical purposes.

(5) We should not expect to eliminate all errors, however. The reason for this is that the Times Atlas gives locations only to the nearest minute of arc, which amounts to roughly 2km. Hence, we should not expect to get a match much better than $\pm$ 1km or so (equivalently, about $\pm\frac{1}{2}$ minute of arc). Since this is about the same order of magnitude as the expected "convergence of the meridians" error, it does not seem appropriate to carry the initial computations beyond the terms of the first degree.

d. We applied the above theory to a few sets of location data, using only the linear terms in the conversion equation. ENCL 4 gives the results of the linear regression fit for the data set that seemed to give the best results. The data set used consists of 19 locations extracted from the ACSDB files, plus an additional 20 locations (each of which is the location of the Nord de Guerre "False Origin," whose location is known by definition). As shown there, the standard error of estimate in longitude amounts to about 0.0060 decimal degrees, and in latitude to about 0.0055 decimal degrees. Since the expected minimum error is about $\pm\frac{1}{2}$ minute of arc, or about 0.0083 decimal degrees, these standard errors of estimate appear to be acceptable.

e. ENCL 5 gives the equations for converting UniGrid coordinates to latitude and longitude, and for the reverse conversion.

7. **Known Errors in the ACSDB.** This paragraph describes the errors that have been found as of this date in the ACSDB's Unit Location File. However, only the START_LOC (i.e., the starting location) field of that file has been examined so far. Also, it should be understood that the review conducted so far would be expected to turn up only the most obvious and striking errors—more subtle errors would have escaped notice. The errors are listed in order by date, where we count D-Day as 15 Dec 1944, D+1 as 16 Dec 1944, etc. In this system, the ACSDB contains information on days from D-Day to D+32, inclusive, as indicated by the following table of correspondences.

Month	Day	D+	Month	Day	D+
Dec	15	0	Jan	01	17
	16	1		02	18
	17	2		03	19
	18	3		04	20
	19	4		05	21
	20	5		06	22
	21	6		07	23
	22	7		08	24
	23	8		09	25
	24	9		10	26
	25	10		11	27
	26	11		12	28
	27	12		13	29
	28	13		14	30
	29	14		15	31
	30	15		16	32
	31	16			

a. On D+1 (16 Dec 44), the German unit recorded in the data base as at P025852 (BROICH) is actually at F025852 (BROICH), since there are other entries for BROICH at those coordinates.

b. On D+4 (19 Dec 44), the German unit recorded in the data base as at P7096 (WANNE) appears to be too far north and west. However, the coordinates P7096 do correspond to 50°21′N, 5°55′E, which *is* the location of Wanne, Belgium. Also, this appears to be the location of KampfGruppe Peiper.

c. On D+4 and D+5 (19-20 Dec 44), the British unit recorded in the data base as at K6665 (GEELEN) appears to be too far east. However, its coordinates do correspond to 50°58′N, 5°58′E, which *is* the location of Geleen, Netherlands, and (except for minor spelling differences) appears to be the place indicated. On the other hand, the locations given in the fields for SIGEV_OTH on 19 Dec 44, and for OBSTACLES, SIGEV_OTH, REFPT1 through REFPT4, and REMARKS for 20 Dec 44, seem to place this unit closer to (say) K3665 or K3130 than to K6665.

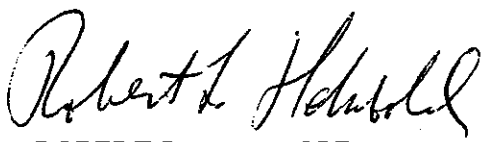
d. On D+10 through D+13 (25-28 Dec 44), there appear to be two British units occupying locations near O8195. The data base records their positions as O809957 (METTET) and O810952. Is O809957 a misprint for O890957? It is difficult to tell.

e. On D+19 through D+26 (3-10 Jan 45), the German unit recorded in the data base as at P408369 (LAVACHERIE) is actually at P408639 (LAVACHERIE), since the other entries for Lavacherie are at P410640 and P408639.

f. On D+21 (5 Jan 45), the German unit recorded in the data base as at P3324 looks wrong. This unit is probably at F3324, since that is more consistent with the information in fields REFPT1 through REFPT4.

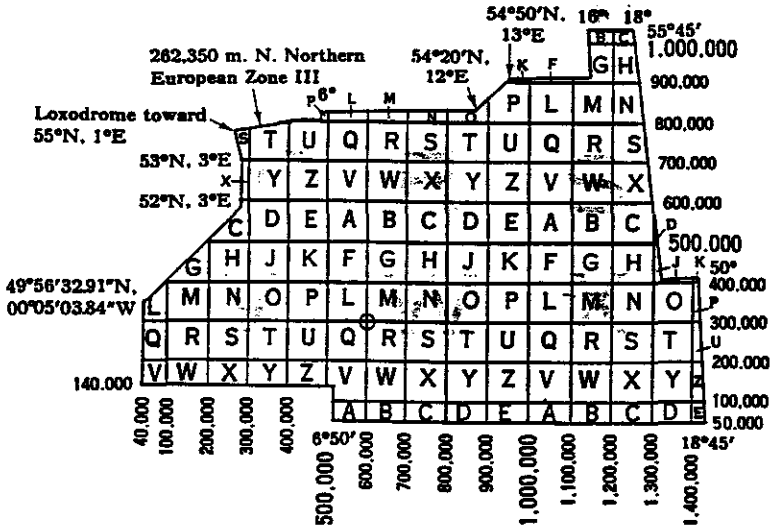
g. On D+24 (8 Jan 45), the unit recorded in the data base as at K467980 (VS GERTRUDE) is actually at P467980 (VS GERTRUDE), since that is the location of Villers-St Gertrude on the map, and since there are other data base entries for VS GERTRUDE at that location.

h. Finally, as noted earlier, the entry k5400 (WORBOMONT) is a typo for K5400 (WORBOMONT).



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NORD DE GUERRE ZONE



PROJECTION: Lambert Conical Orthomorphic (modified)

SPHEROID: Du Plessis (reconstituted)

UNIT: Meter

ORIGIN: 49°30'N., 7°44'13.95"E.

FALSE COORDINATES OF ORIGIN: 600,000 meters E.; 300,000 meters N.

SCALE FACTOR: 0.999509082

INCIDENCE OF GRID LETTERS: The 500,000-meter square letter V and the 100,000-meter square letter A are both north and east of the false origin.

GRID TABLES: *Lambert Conical Orthomorphic Projection Tables, Nord de Guerre Zone*, AMS (Key No. 200522)

GRID "COLOR": Blue

REFERENCING FOR 1,000-METER GRID (6-digit numerical reference):
Principal digits: (2); 10,000, 1,000
Reference box: AMS Cut No. 160. Modifications to box: None

REFERENCING FOR 10,000-METER GRID (4-digit numerical reference):
Principal digits: (1); 10,000
Reference box: AMS Cut No. 162. Modifications to box: None

ENCL 9

17 OCT 1991

ACSDB SEQNO	ACSDB Year	ACSDB MoDay	ACSDB Start_Loc	Map Grid Easting	Map Grid Northing	Latitude Deg.Min	Longitude Deg.Min	Long DecDeg	Calc'd Long	Error Long	Lat DecDeg
50	1944	1217	F1115 (SCHLEIDEN)	11.0	15.0	50.32	6.29	6.483	6.480	-0.0037	50.533
74	1944	1229	F1220 (GEMUEND)	12.0	20.0	50.34	6.30	6.500	6.493	-0.0074	50.567
74	1944	1216	F165027(SCHMIDTHEIM)	16.5	2.7	50.26	6.34	6.567	6.558	-0.0083	50.433
35	1944	1216	F2251 (BLATZHEIM)	22.0	51.0	50.51	6.38	6.633	6.626	-0.0076	50.850
17	1944	1219	F230048(BLANKENHEIM)	23.0	4.8	50.26	6.39	6.650	6.648	-0.0017	50.433
71	1944	1216	F3213 (SCHOENAU)	32.0	13.0	50.30	6.47	6.783	6.772	-0.0115	50.500
42	1944	1227	K798270 (EUPEN)	79.8	27.0	50.38	6.02	6.033	6.044	0.0106	50.633
49	1944	1227	K8353 (KERKRADE)	83.0	53.0	50.52	6.04	6.067	6.084	0.0169	50.867
23	1944	1215	K910041 (BUTGENBACH)	91.0	4.1	50.26	6.12	6.200	6.204	0.0039	50.433
17	1944	1215	K918293 (ROTGEN)	91.8	29.3	50.38	6.13	6.217	6.210	-0.0064	50.633
3	1944	1215	K977566 (ALDENHOVEN)	97.7	56.6	50.54	6.17	6.283	6.287	0.0038	50.900
56	1944	1223	L065795 (PRUEM)	6.5	79.5	50.12	6.25	6.417	6.424	0.0071	50.200
34	1944	1229	P215428 (BERTRIX)	21.5	42.8	49.52	5.15	5.250	5.250	-0.0002	49.867
102	1944	1223	P310610 (ST. HUBERT)	31.0	61.0	50.02	5.22	5.367	5.378	0.0117	50.033
3	1945	0103	P3440 (NEUFCHATEAU)	34.0	40.0	49.51	5.26	5.433	5.424	-0.0093	49.850
29	1944	1215	P5558 (BASTOGNE)	55.0	58.0	50.00	5.43	5.717	5.712	-0.0043	50.000
3	1944	1223	P670317 (REDANGE)	67.0	31.7	49.46	5.53	5.883	5.884	0.0007	49.767
7	1944	1215	P700530 (WILTZ)	70.0	53.0	49.58	5.56	5.933	5.922	-0.0116	49.967
36	1945	0114	P9783 (BLEIALF)	97.0	83.0	50.14	6.17	6.283	6.291	0.0078	50.233
1	---	---	False Origin	200.0	0.0			7.737208	7.738	0.0005	49.500
2	---	---	False Origin	200.0	0.0			7.737208	7.738	0.0005	49.500
3	---	---	False Origin	200.0	0.0			7.737208	7.738	0.0005	49.500
4	---	---	False Origin	200.0	0.0			7.737208	7.738	0.0005	49.500
5	---	---	False Origin	200.0	0.0			7.737208	7.738	0.0005	49.500
6	---	---	False Origin	200.0	0.0			7.737208	7.738	0.0005	49.500
7	---	---	False Origin	200.0	0.0			7.737208	7.738	0.0005	49.500
8	---	---	False Origin	200.0	0.0			7.737208	7.738	0.0005	49.500
9	---	---	False Origin	200.0	0.0			7.737208	7.738	0.0005	49.500
10	---	---	False Origin	200.0	0.0			7.737208	7.738	0.0005	49.500
11	---	---	False Origin	200.0	0.0			7.737208	7.738	0.0005	49.500
12	---	---	False Origin	200.0	0.0			7.737208	7.738	0.0005	49.500
13	---	---	False Origin	200.0	0.0			7.737208	7.738	0.0005	49.500
14	---	---	False Origin	200.0	0.0			7.737208	7.738	0.0005	49.500
15	---	---	False Origin	200.0	0.0			7.737208	7.738	0.0005	49.500
16	---	---	False Origin	200.0	0.0			7.737208	7.738	0.0005	49.500
17	---	---	False Origin	200.0	0.0			7.737208	7.738	0.0005	49.500
18	---	---	False Origin	200.0	0.0			7.737208	7.738	0.0005	49.500
19	---	---	False Origin	200.0	0.0			7.737208	7.738	0.0005	49.500
20	---	---	False Origin	200.0	0.0			7.737208	7.738	0.0005	49.500
							Min	5.250	5.250	-0.012	49.500
							Max	7.737	7.738	0.017	50.900
							Avg	6.960	6.960	-0.000	49.915

Longitude (DecDeg) Regression Output:

Constant	4.9591431172831
Std Err of Y Est	0.005996135979496
R Squared	0.99995451963263
No. of Observations	39
Degrees of Freedom	36
Coefficient(s) 0.013892742153339	-0.00018761339619042
Std Err of Coef. 2.177391038121E-05	2.4090606236274E-05

Latitude (DecDeg) Regression Output:

Constant	49.480490564919
Std Err of Y Est	0.0055366770314816
R Squared	0.999882809116
No. of Observations	39
Degrees of Freedom	36
Coefficient(s) 9.7296500661E-05	0.0089903810302589
Std Err of Coef. 2.010546623782E-05	2.224464333013E-05

UniGrid Easting (km)	UniGrid Northing (km)	Computed Long (DecDeg)	Computed Lat (DecDeg)	Computed Long (Deg.Min)	Computed Lat (Deg.Min)
0	0	4.95914	49.48049	4.57549	49.28829
0	100	4.94038	50.37953	4.56423	50.22772
0	200	4.92162	51.27857	4.55297	51.16714
100	0	6.34842	49.49022	6.20905	49.29413
100	100	6.32966	50.38926	6.19779	50.23355
100	200	6.31089	51.28830	6.18654	51.17298
200	0	7.73769	49.49995	7.44261	49.29997
200	100	7.71893	50.39899	7.43136	50.23939
200	200	7.70017	51.29803	7.42010	51.17882

ENCL 4

alc'd Lat	Error Lat	UniGrid Easting	UniGrid Northing	Map Sheet	Add km Easting	Add km Northing
50.525	-0.0081	111.0	115.0	F	100	100
50.570	0.0036	112.0	120.0	F	100	100
50.415	-0.0182	116.5	102.7	F	100	100
50.850	-0.0001	122.0	151.0	F	100	100
50.435	0.0013	123.0	104.8	F	100	100
50.509	0.0092	132.0	113.0	F	100	100
50.630	-0.0033	79.8	127.0	K	0	100
50.864	-0.0026	83.0	153.0	K	0	100
50.425	-0.0081	91.0	104.1	K	0	100
50.652	0.0185	91.8	129.3	K	0	100
50.898	-0.0021	97.7	156.6	K	0	100
50.206	0.0056	106.5	79.5	L	100	0
49.867	0.0007	21.5	42.8	P	0	0
50.032	-0.0014	31.0	61.0	P	0	0
49.843	-0.0066	34.0	40.0	P	0	0
50.007	0.0073	55.0	58.0	P	0	0
49.772	0.0053	67.0	31.7	P	0	0
49.964	-0.0029	70.0	53.0	P	0	0
50.236	0.0028	97.0	83.0	P	0	0
49.500	-0.0001	200.0	0.0	M	200

ENCLOSURE 5

EQUATIONS FOR CONVERTING UNIGRID COORDINATES TO AND FROM LAT/LONG

5-1. Based on the computations shown at ENCL 4, the equations relating the UniGrid and Lat/Long coordinate systems are estimated to be as follows. I believe they provide a level of accuracy sufficient for most CAA purposes.

$$\begin{bmatrix} \text{Long} \\ \text{Lat} \end{bmatrix} = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} + M \begin{bmatrix} \text{UGE} \\ \text{UGN} \end{bmatrix} ,$$

where

$$\begin{bmatrix} B_1 \\ B_2 \end{bmatrix} = \begin{bmatrix} 4.9591431172831 \\ 49.480490564919 \end{bmatrix} ,$$

$$M = \begin{bmatrix} 0.013892742153339 & -0.00018761339619042 \\ 9.7296500661\text{E}-05 & 0.0089903810302589 \end{bmatrix} .$$

In these equations, UGE and UGN are the UniGrid easting and northing coordinates of a point (in kilometers) and Long(itude E) and Lat(itude N) are expressed in degrees and decimal fractions of a degree. The components of the vector $[B_1, B_2]^T$ are the estimated Long and Lat of the origin of of the UniGrid coordinate system.

5-2. The inverse relationships are:

$$\begin{bmatrix} \text{UGE} \\ \text{UGN} \end{bmatrix} = M^{-1} \begin{bmatrix} \text{Long} - B_1 \\ \text{Lat} - B_2 \end{bmatrix} ,$$

where M^{-1}, the matrix inverse to M, is given by:

$$M^{-1} = \begin{bmatrix} 71.9695119659326 & 1.501876785494469 \\ -0.7788748491301217 & 111.2137371351828 \end{bmatrix} .$$

In these (inverse) equations, Lat and Long are to be expressed in degrees and decimal fractions of a degree, and UGE and UGN are the UniGrid easting and northing in km.